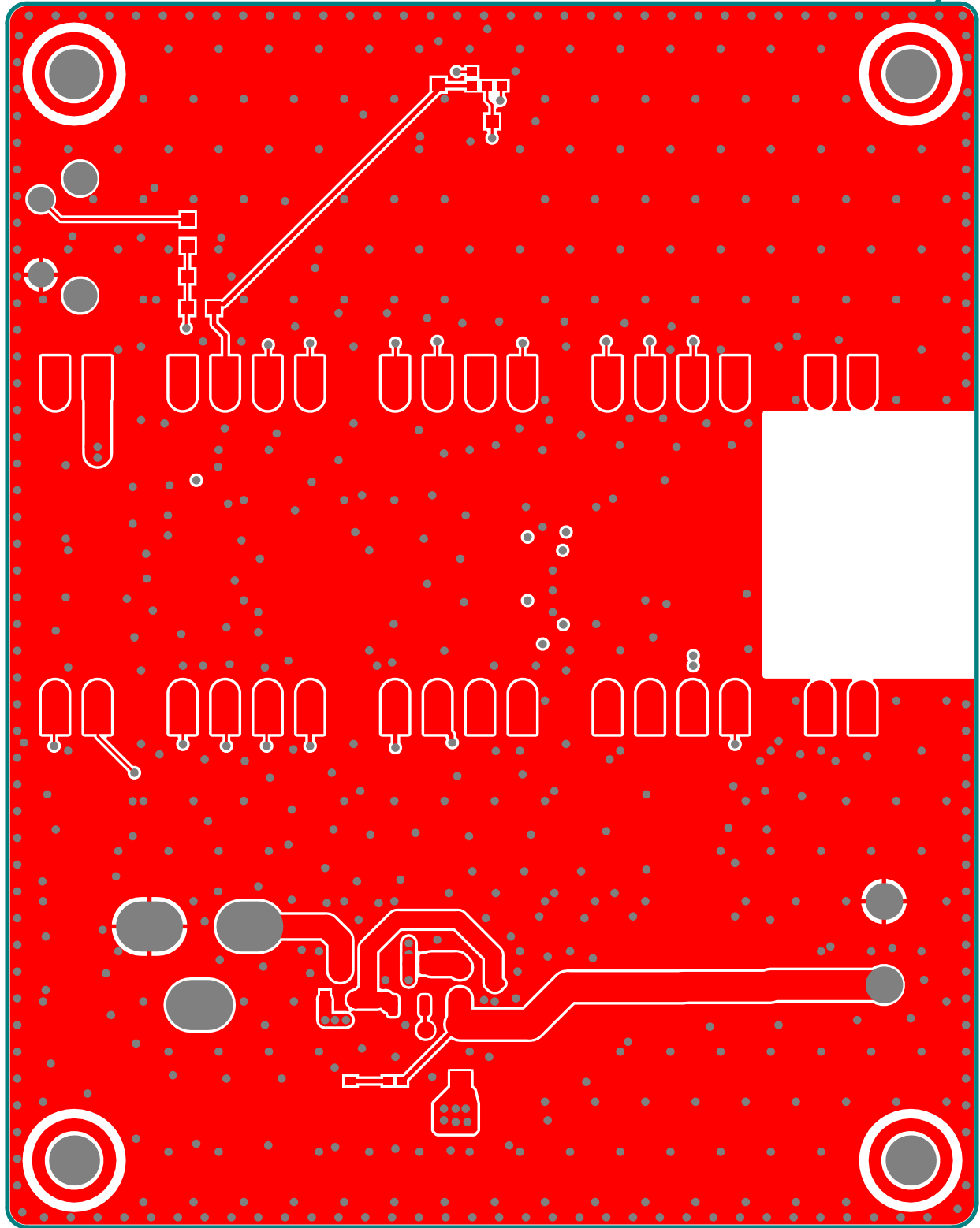
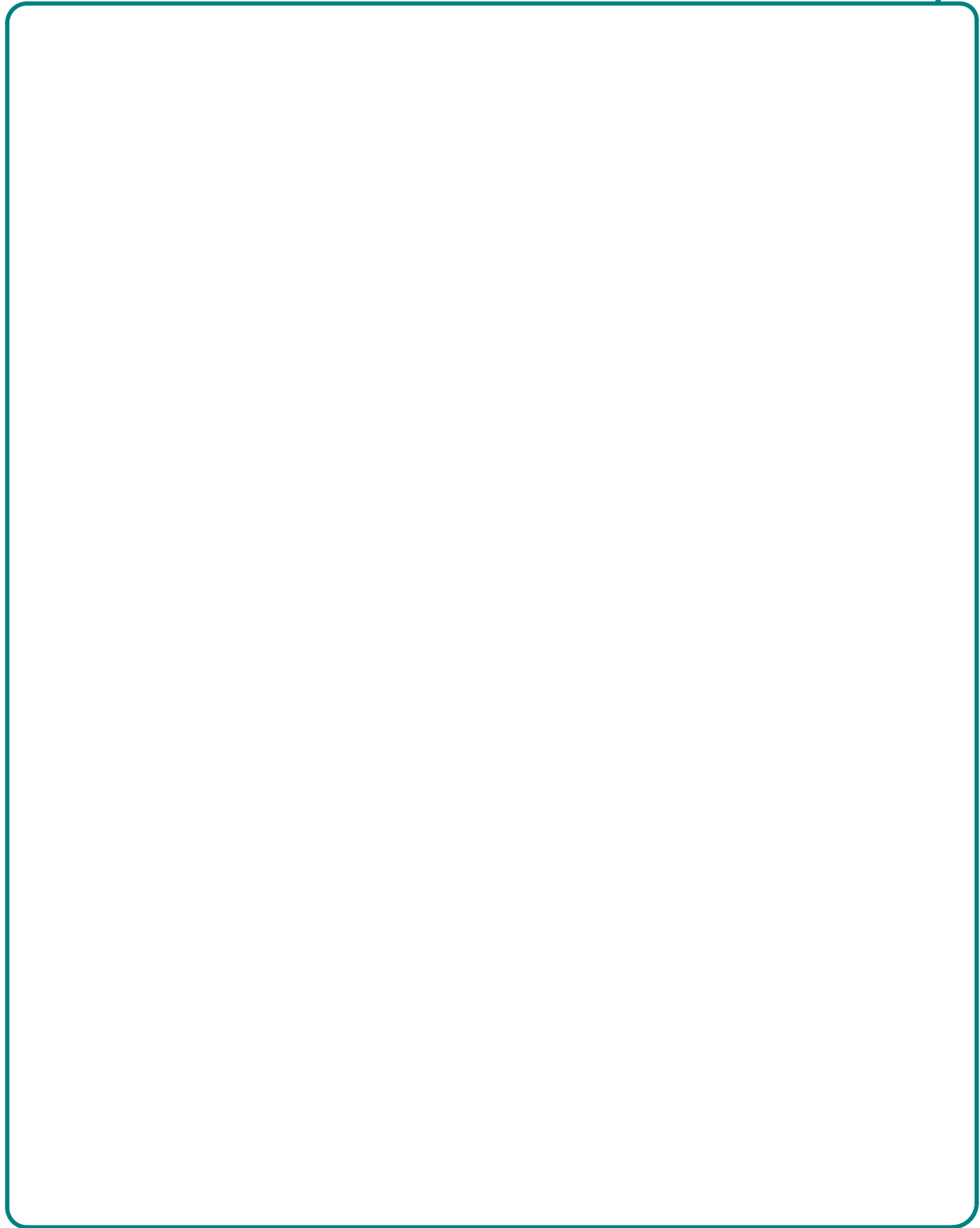
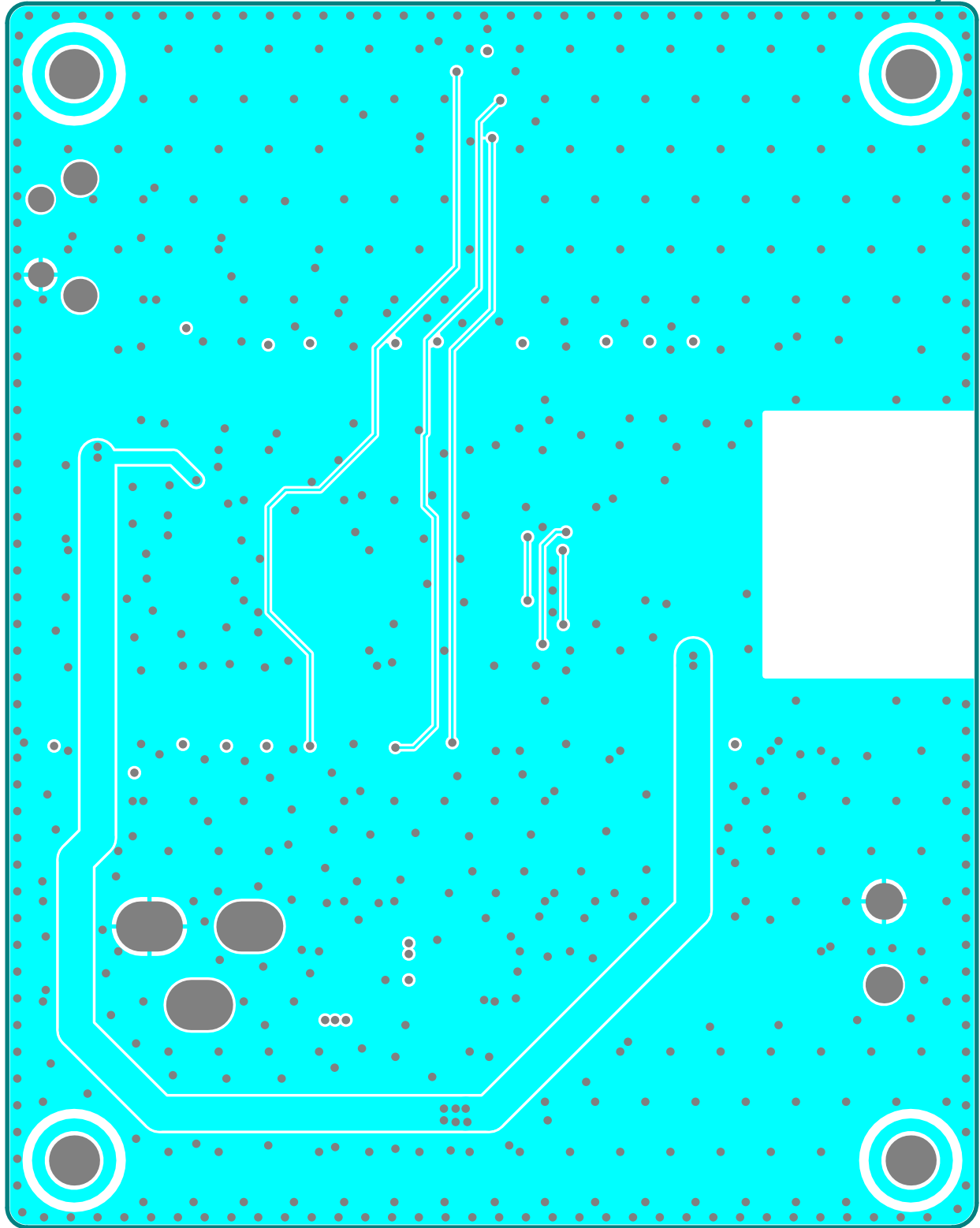
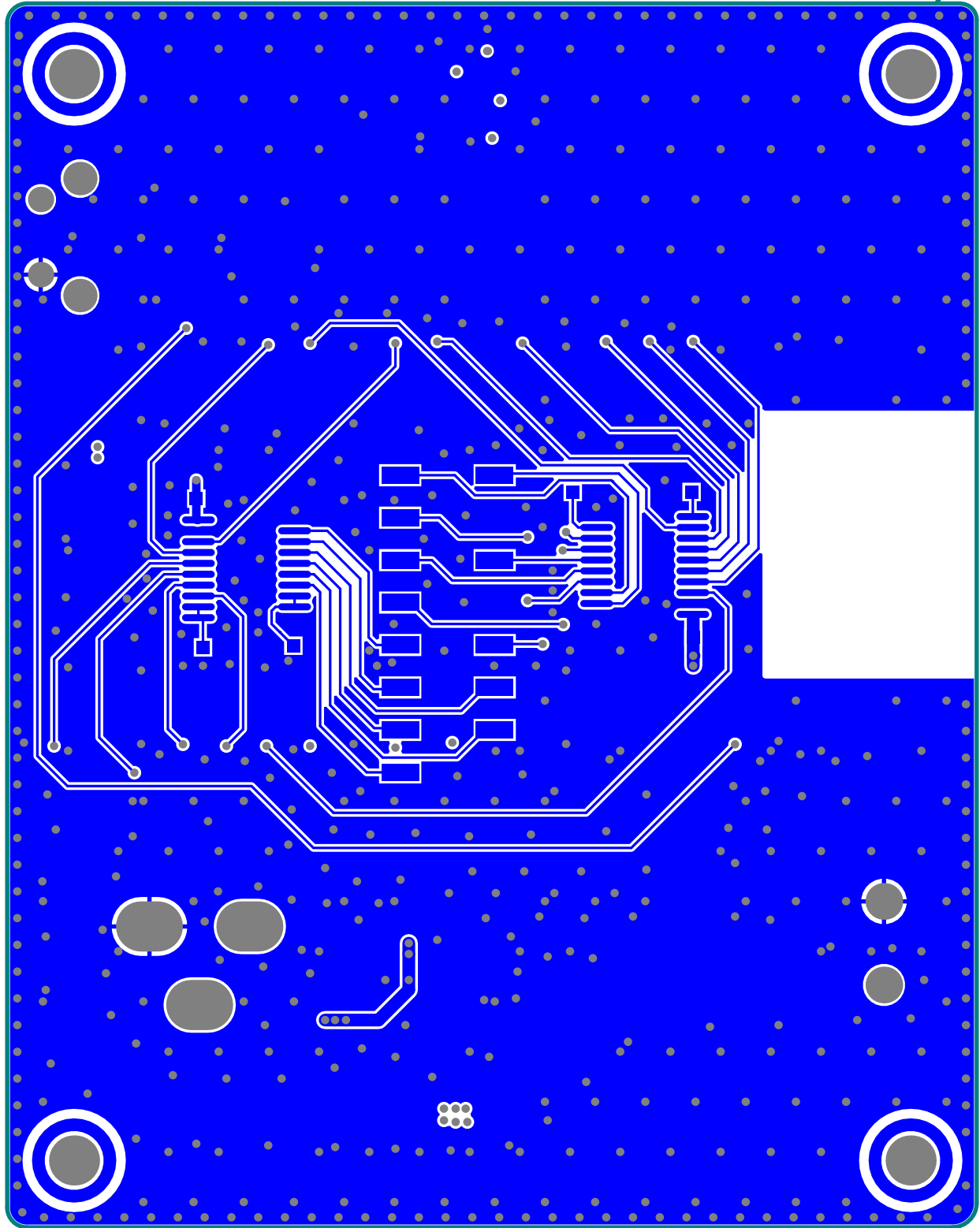
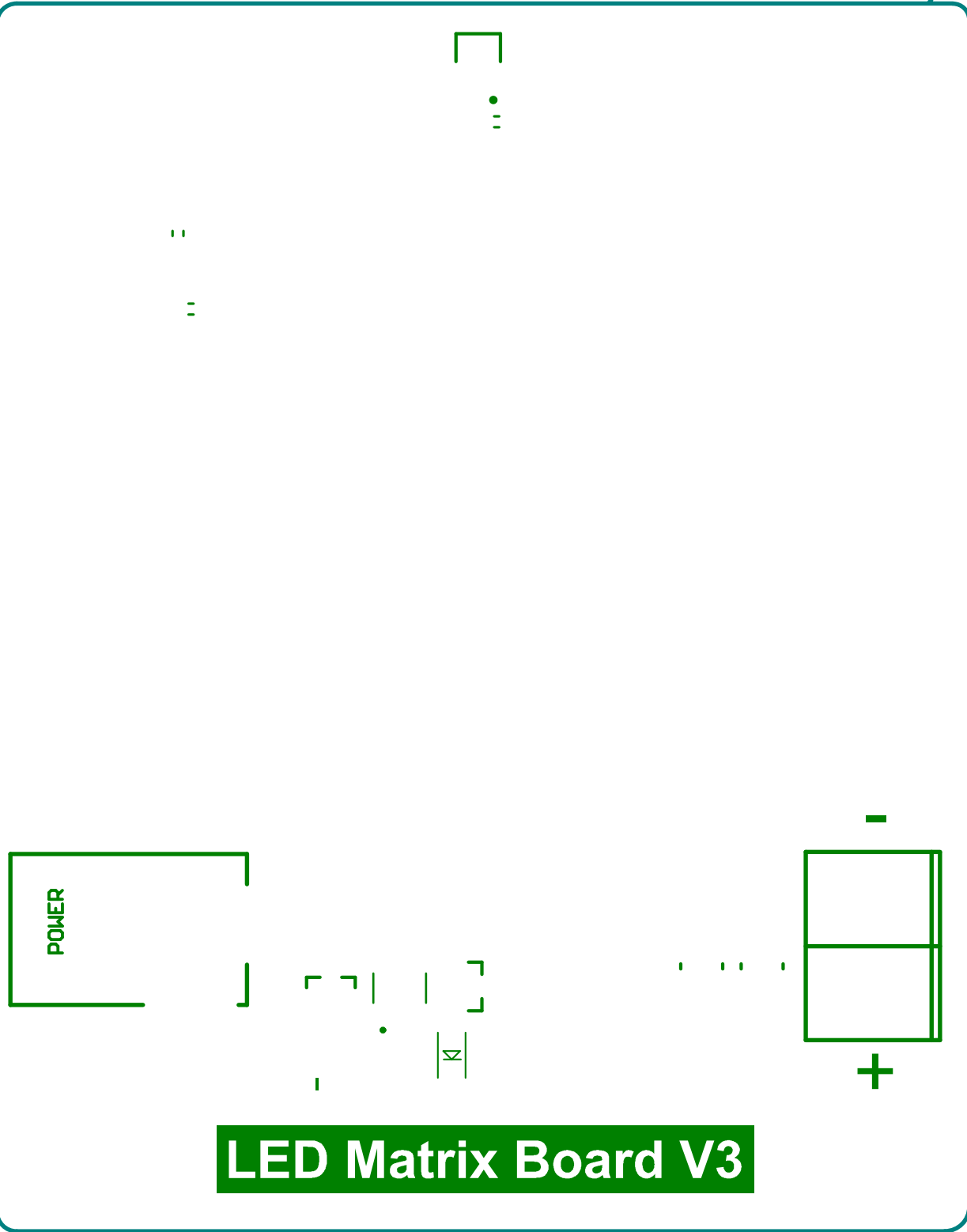




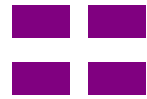
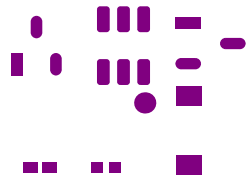


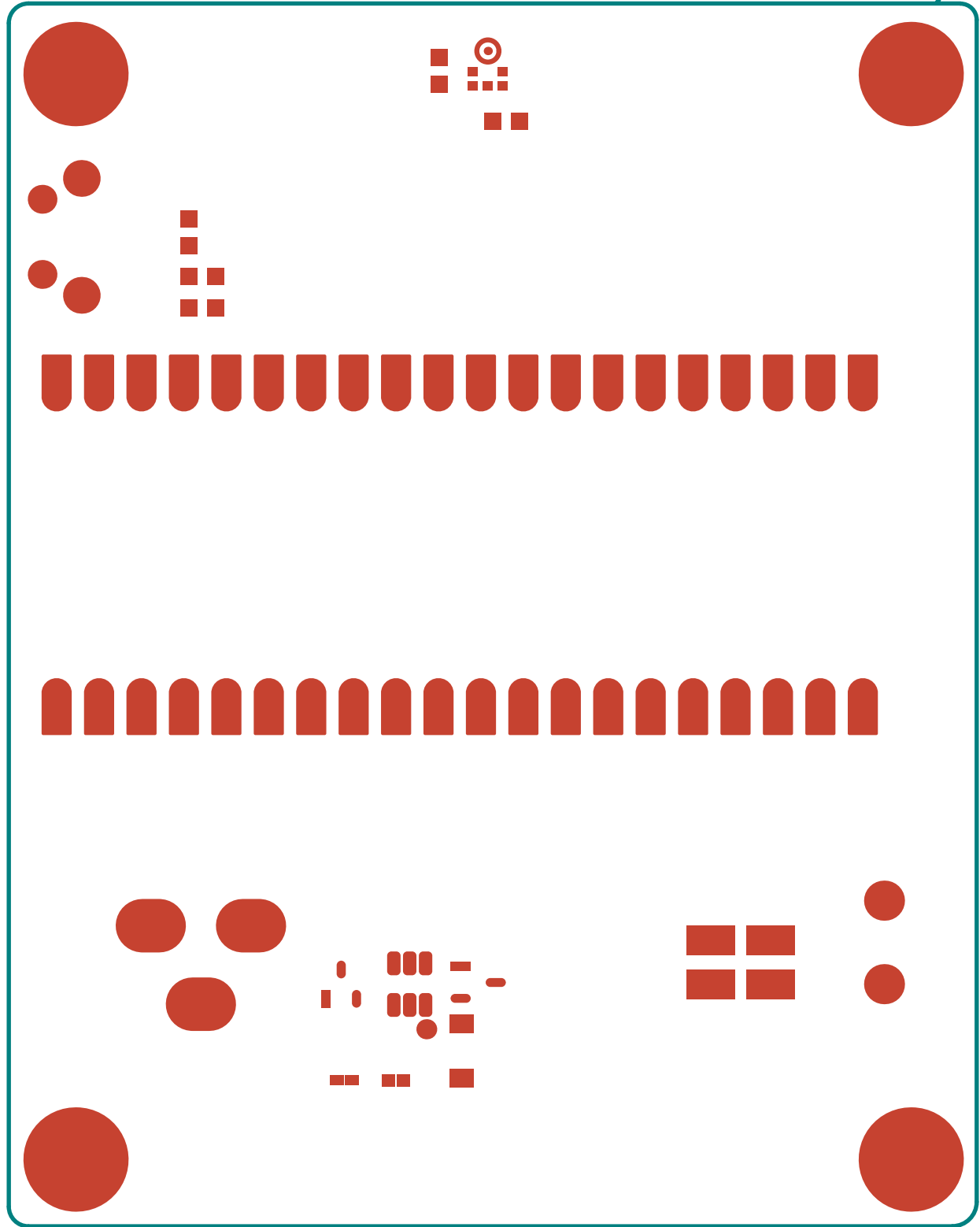


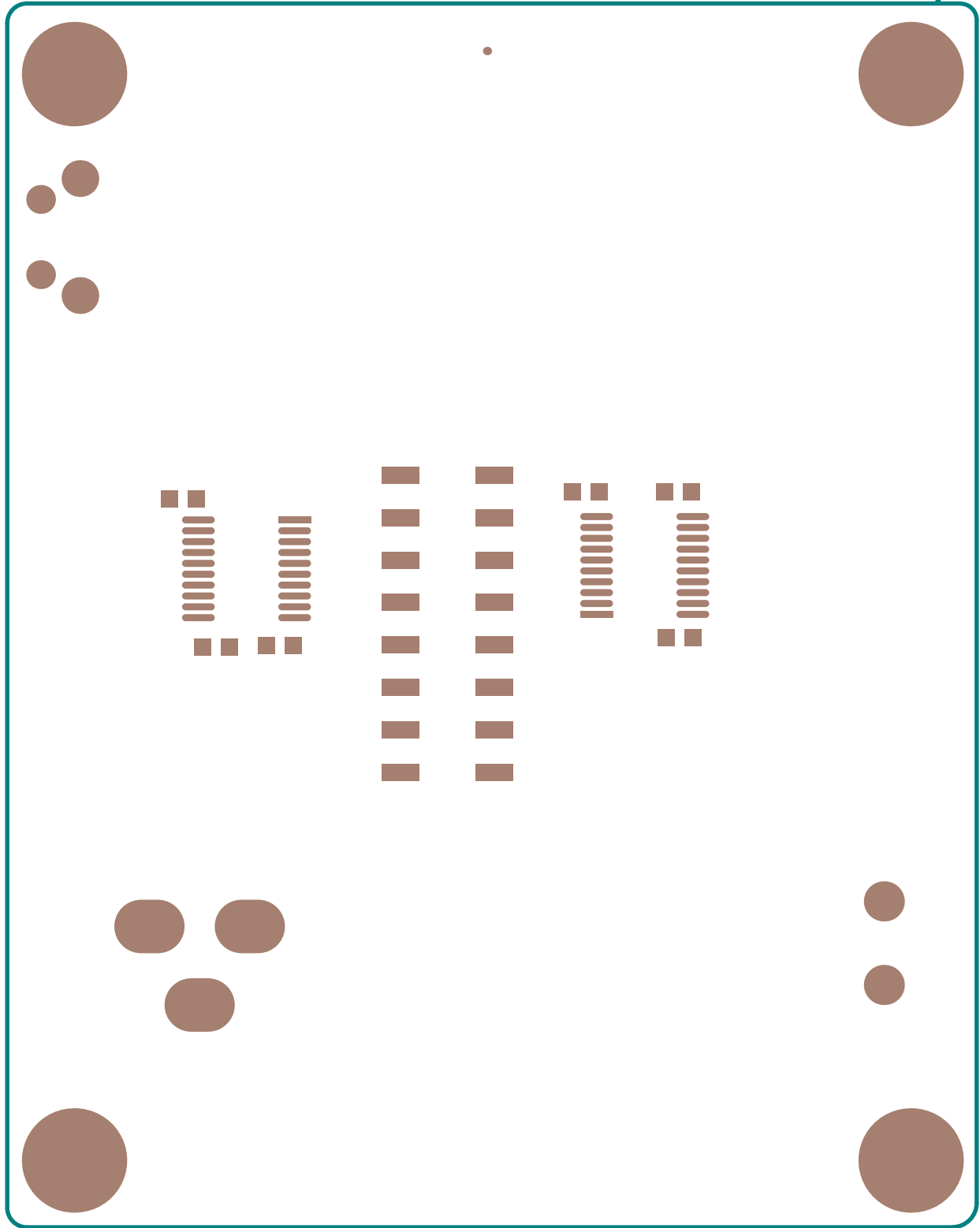


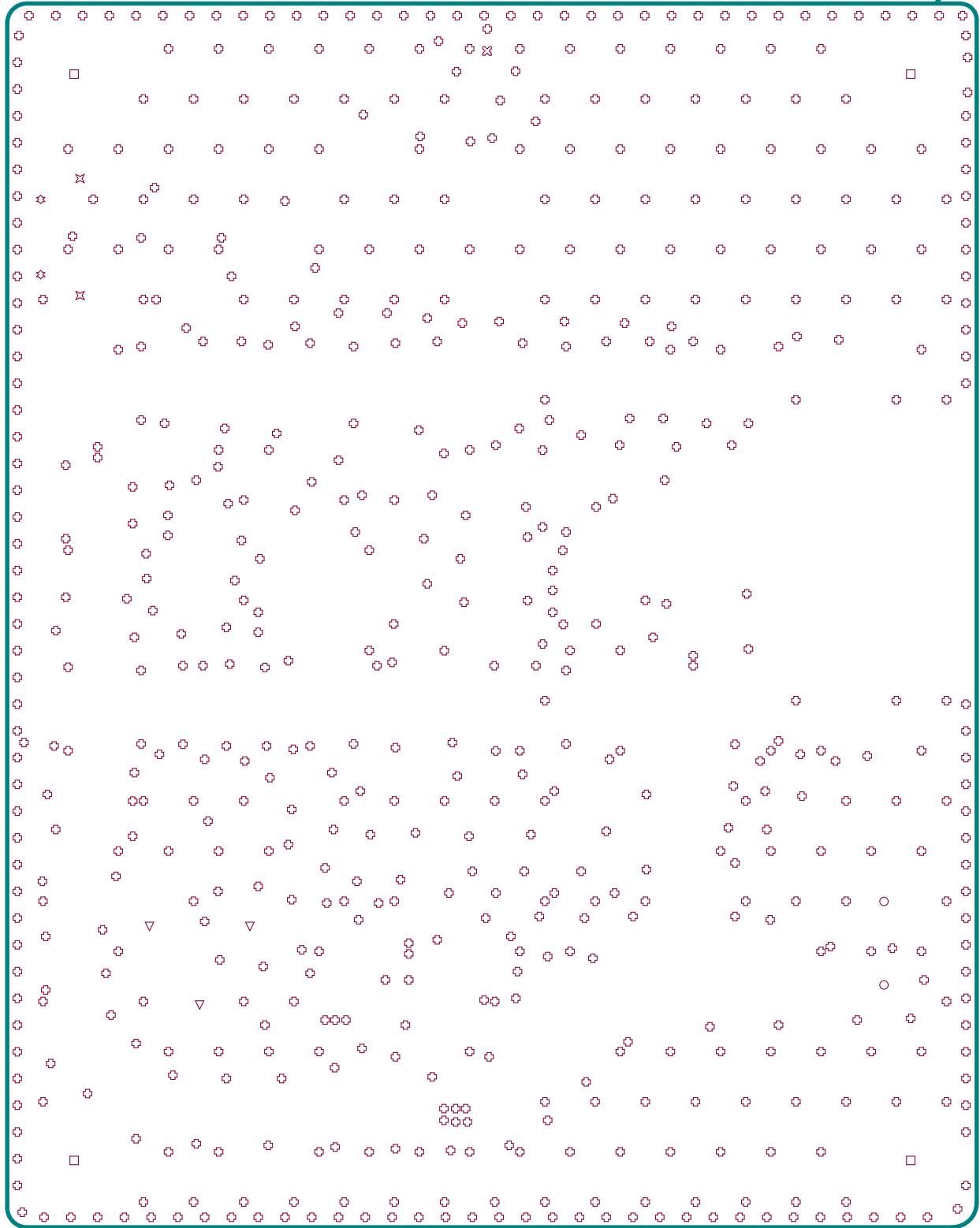


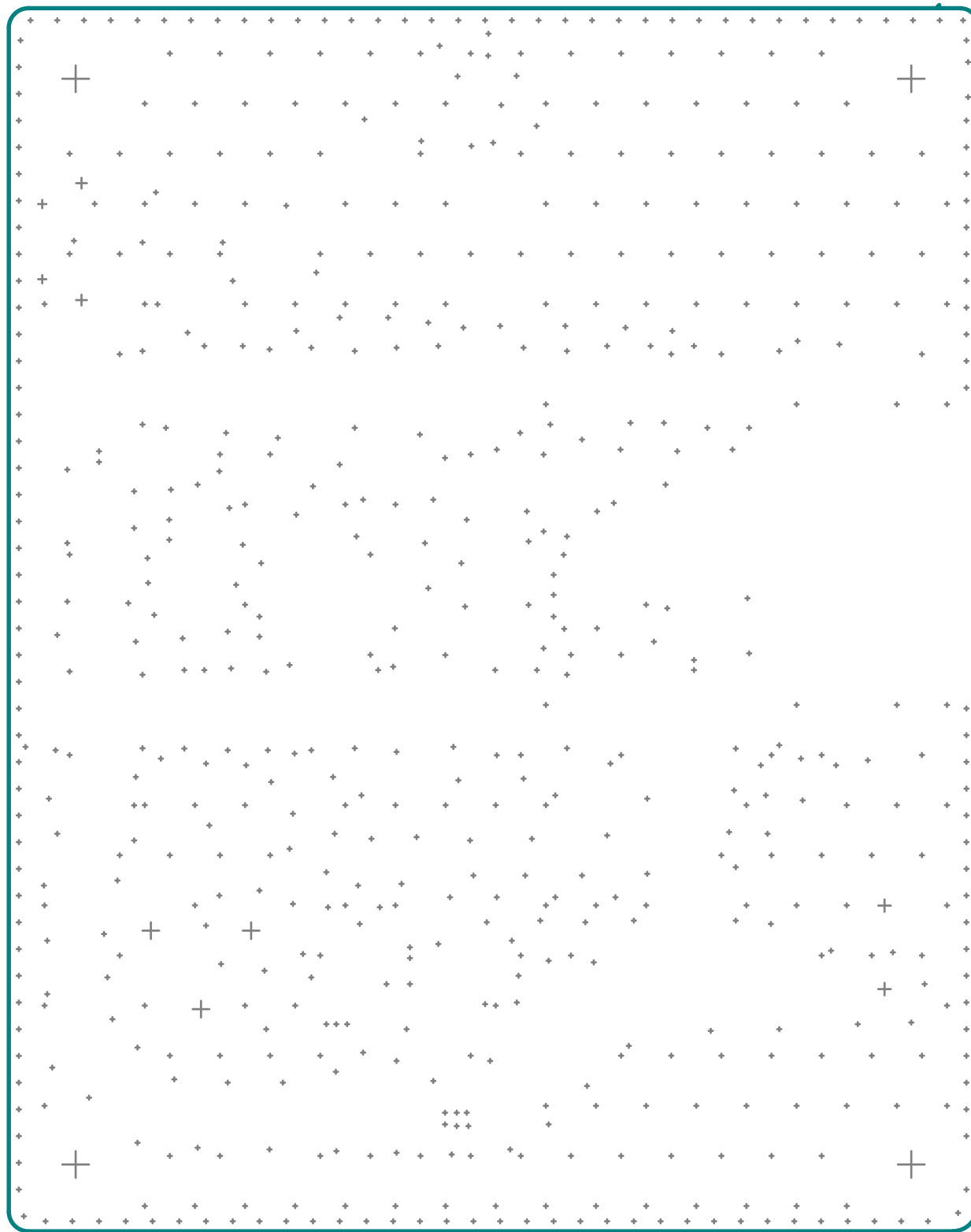
LED Matrix Board V3











NOTES: UNLESS OTHERWISE SPECIFIED0

1. SPECIFICATIONS/TOLERANCES:

- A. FABRICATE PER IPC-6012, CLASS 2, USING PROVIDED DATA FILES XXXXXXXX.ZIP
- B. ALL SPECIFICATIONS USED SHALL BE PER THEIR LATEST REVISIONS.
- C. THE DIMENSIONS OF CIRCUIT FEATURES IN THE PROVIDED DATA MAY BE ADJUSTED ONLY TO COMPENSATE FOR PROCESS TOLERANCES; ADDING, REMOVING OR RELOCATING CIRCUIT FEATURES IS NOT ALLOWED
- D. REMOVE ALL BURRS AND BREAK SHARP EDGES, .381 [.015] MAX RADIUS.
- E. PARENTHETICAL INFORMATION IS FOR REFERENCE ONLY.
- F. REPAIR OF PCB DEFECTS IS NOT PERMITTED.

2. DIELECTRIC MATERIAL:

- A. DIELECTRIC MATERIAL SHALL BE PER IPC-4101/99, /124, /126 OR /129 (RoHS COMPLIANT EPOXY-GLASS)
- B. MINIMUM DIELECTRIC THICKNESS SHALL BE .051 [.002] FOR REFERENCED STACK-UP DIMENSIONS OF.076 [.003] OR GREATER; IPC-6012 REQUIREMENTS SHALL OTHERWISE APPLY. SINGLE-PLY CONSTRUCTION IS ALLOWED.
- C. SEE LAYER STACK-UP FOR REQUIRED COPPER WEIGHTS AND THE FINISHED PCB THICKNESS. IF SPECIFIED, 1/3 OZ. STARTING FOIL MAY BE ACHIEVED BY 1/2 OZ. FOIL REDUCTION.
- D. FINISHED PCB THICKNESS SHALL BE MEASURED OVER LANDS AND/OR CONDUCTORS NOT COVERED BY SOLDER MASK. IF SPECIFIED, 1/3 OZ. STARTING FOIL MAY BE ACHIEVED BY 1/2 OZ. FOIL REDUCTION.

3. DRILLING:

- A. VIA DIAMETERS SHALL BE VERIFIED BEFORE PLATING; ALL OTHER HOLE DIAMETERS ARE FINISHED DIMENSIONS.
- B. LAYER-TO-LAYER MISREGISTRATION SHALL BE .127 [.005] MAXIMUM.

4. SOLDER MASK:

- A. APPLY LPI SOLDER MASK OVER BARE COPPER USING PROVIDED DATA.
- B. SOLDER MASK SHALL BE PER IPC-SM-840, CLASS T, COLOR PURPLE.
- C. THE DIMENSIONS OF SOLDER MASK-DEFINED PADS ON PLANES AND/OR WIDE CONDUCTORS SHALL NOT BE MODIFIED.

5. SILKSCREEN/MARKING:

- A. SILKSCREEN PCB PER PROVIDED DATA USING PERMANENT, NON-CONDUCTIVE INK, COLOR WHITE.
- B. SUPPLIER ID AND TRACEABILITY INFORMATION SHALL NOT BE APPLIED ON ANY PART OF THE PCB
- C. INK SHALL NOT BE APPLIED TO ANY SOLDERABLE SURFACE. USING PROVIDED DATA.

6. ELECTRICAL TEST:

- A. DESIGN VERIFICATION SHALL BE DONE PRIOR TO PCB FABRICATION USING GERBER DATA AND AN IPC-D-356 NETLIST.
- B. ALL PCBs SHALL BE 100% ELECTRICALLY TESTED FOR OPENS AND SHORTS
- C. APPLY TEST STAMP IN NON-LEGEND AREA ON SECONDARY SIDE OF PCB.

7. FINAL FINISH:

- A. FINAL FINISH SHALL BE ELECTROLESS NICKEL/IMMERSION GOLD (ENIG) PER IPC-4552.

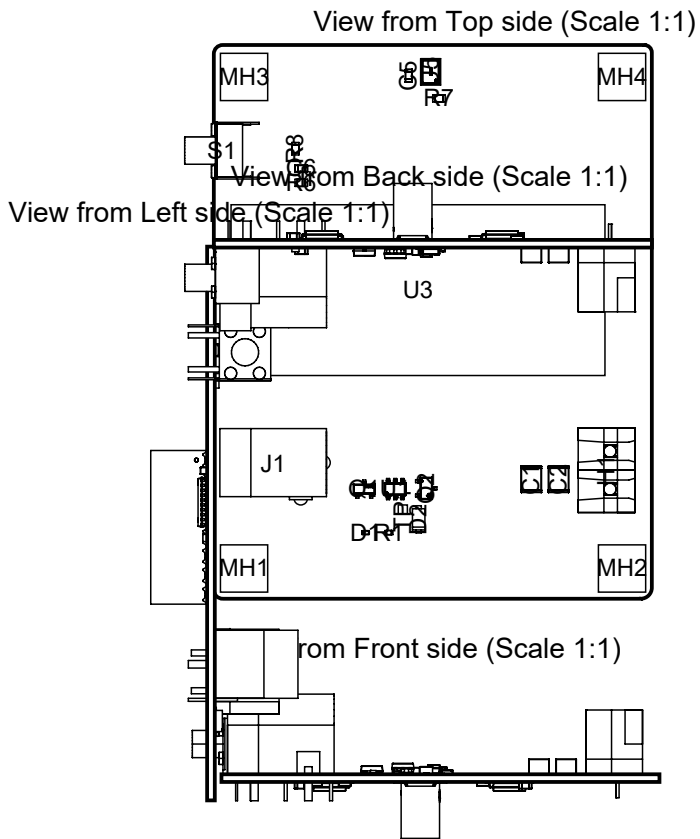
8. IMPEDANCE:

- A. IMPEDANCE TOLERANCE SHALL BE +/- 10%.
- B. SEE LAYER STACK-UP FOR IMPEDANCE REQUIREMENTS.

- 9. IF PANELIZATION SPECIFICATIONS ARE PROVIDED, THE PCBs SHALL BE DELIVERED IN PANEL FORM. HOWEVER, THESE SPECIFICATIONS MAY BE REPLACED BY SUPPLIER PREFERRED (SPECIFICATIONS AS APPROVED).



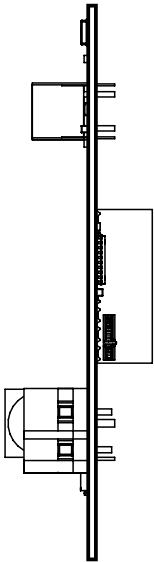
	Project: LED Matrix Board	
	Company:	
	Drawn By: Jonathan Garnier	
	Date: 6/05/2026	Rev: 3.0



Transmission Line Structure Table

Transmission Line	Target Impedance	Calculated Impedance	Trace layer	Wide Trace Width	Reference layers
Coated Coplanar Waveguide With Ground	50	13.35	Top Layer	0.80mm	Layer 1
Embedded Coplanar Waveguide	50	0.00	Layer 2	0.00mm	

View from Right side (Scale 1:1)



Layer Stack Legend

Material	Layer	Thickness	Dielectric Material	Type	Gerber
Top Overlay	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.02mm	PSR-4000-BN-Colour	Solder Mask	GTS
ENIG	Top Surface Finish	0.02mm		Surface Finish	
CF-004	Top Layer	0.04mm		Signal	GTL
Prepreg		0.07mm	PP-006	Dielectric	
CF-004	Layer 1	0.04mm		Signal	G1
Core		0.69mm	Core	Dielectric	
CF-004	Layer 2	0.04mm		Signal	G2
Prepreg		0.07mm	PP-006	Dielectric	
CF-004	Bottom Layer	0.04mm		Signal	GBL
ENIG	Bottom Surface Finish	0.02mm		Surface Finish	
Surface Material	Bottom Solder	0.02mm	PSR-4000-BN-Colour	Solder Mask	GBS
Bottom Overlay	Bottom Overlay			Legend	GBO

Total thickness: 1.05mm